

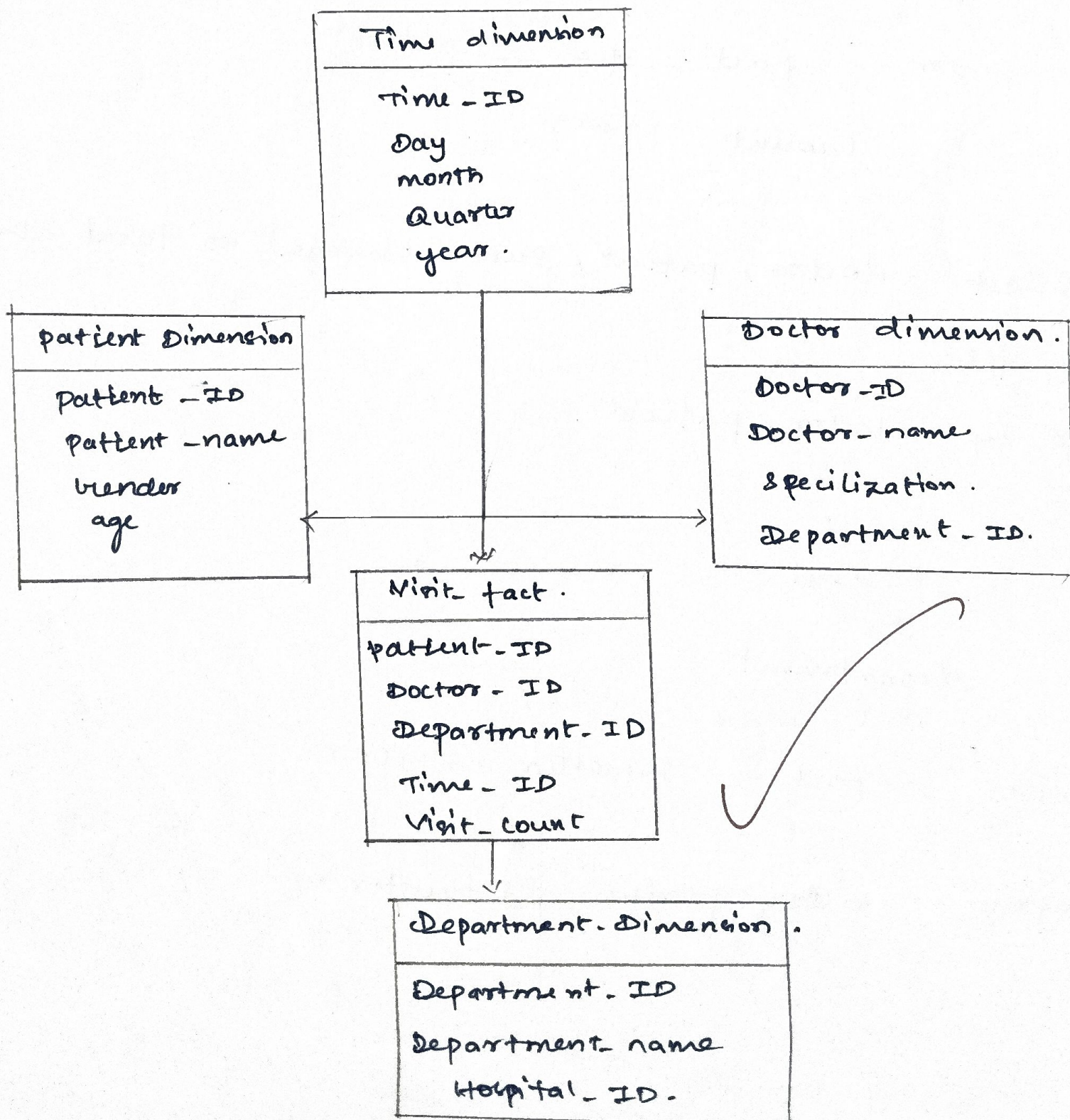
1. City hospital Data Warehouse.

Given dimension .

- 1) patient -
- 2) Doctor -
- 3) Department -
- 4) Time.

Measures.

- 1) Visit - count .
- 2) avg - charge.



2, a) Three popular datawarehouse house Schemes.

Star Schema.

Snowflake Schema.

Galaxy Schema

b, Star Schema.

Time

(Day, month, year)

1

Doctor --- fact --- fee --- patient

(count, charge)

c) select doctor, patient, sum (charge) as total charge
from fee.

group by doctor, patient.

3, a)

Student-

1

department

1

course --- fact --- semester --- year.

1

Program

1

Avg grade

1

Instructor.

}

OLAP operation.

Number of cuboids.

$$= 5 \times 5 \times 5 \times 5 = 625 \text{ cuboids.}$$

Star Schema.

fact table.

Weather - fact

- Air pressure

- temperature

- precipitation.

Dimension table.

- Time (hour $\rightarrow$ Day $\rightarrow$ month $\rightarrow$ year)

- Location (probe $\rightarrow$ region $\rightarrow$ land)

- Weather type (sunny, rainy, cloudy).

5) a) maximum cell in base cuboid.

$$= p^n.$$

b) minimum cells in base cuboid.

$$= p.$$

c) Maximum cell in entire cube.

$$= (p+1)^n$$

d) minimum cells in entire cube.

$$= 2^n.$$